

Diagnostic Procedures and Decision Logic

Robot Weld 01 (robot-weld-01) | Sensor 1: AXIS_4_SERVO_TORQUE

Machine	Robot Weld 01 (machine ID robot-weld-01, six-axis spot-welding robot, Body shop, site vit-lab)
Sensor	Sensor 1: AXIS_4_SERVO_TORQUE, "Axis 4 servo torque", sensor ID ROBOT-WELD-01.AXIS_4_SERVO_TORQUE, type torque, unit Nm
Component / subsystem	Robot forearm (axis 4): axis 4 servo drive and gearbox; planned agent component tag robot_forearm_axis_4; suggested twin mesh 105-six-axis-welding-robot/arm-swing
Document title	Diagnostic Procedures and Decision Logic
Document ID	RW01-S1-DOC-03
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Purpose	Gives the diagnostic agent a structured procedure for interpreting Sensor 1 (axis 4 servo torque) telemetry and deciding what happens next: what happened, how abnormal it is, what could cause it, what should be checked and what action should be recommended.
Scope	Procedure, IF/THEN rules, decision matrices, scoring and escalation for Sensor 1 of Robot Weld 01. Numerical levels come from Document 01, signatures from Document 02; maintenance steps are in Document 04 and worked cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_SENSOR1.md (catalog limits, simulation equations, topics, tests).
[D]	Derived value: computed by the document authors from the repository's own simulation equations (noise-free unless stated).
[E]	Engineering reference: general industrial practice for robot servo-axis diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. An ESP32 gateway (or a software simulator) plays the robot controller and generates values from documented equations. The limits in this document that are tagged [P] are illustrative constants chosen for the demo in the repository, not OEM specifications. Values tagged [E] and [I] are not manufacturer-certified.

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1. Diagnostic Prerequisites

1.1 Inputs the agent must have

Table 1. Diagnostic prerequisites

Prerequisite	Requirement	Source	If missing
Telemetry window	At least 5 cycles (40 s, 80 samples) of AXIS_4_SERVO_TORQUE for features; 1 cycle (16 samples) minimum for a preliminary result	telemetry topic or GET /telemetry/history [P]	State that the window is too short; give preliminary result only
Device state	status online; not stale; synced true on the samples used	status topic, device registry [P]	Report data-path fault (F12) before any mechanical diagnosis
Sequence continuity	Fewer than 10 percent missing samples in window; no bootId change inside window	seq, bootId [P]	Mark window incomplete
Operating state	Production cycle at steady speed, warm-up complete	Program, speed override, time since start [E]	Do not apply thresholds; ask for context
Baseline	Healthy baseline features (M, template, sigma, A6)	Baseline procedure, Document 01 Section 13	Use simulator defaults (M 13.0, sigma 0.45, A6 0) and flag as assumed [D]
Companion channels	TOOL_CENTER_POINT_DEVIATION and WELD_GUN_TEMP for the same window	telemetry [P]	Mark corroboration unavailable and reduce confidence
Binding	Twin binding of ROBOT-WELD-01.AXIS_4_SERVO_TORQUE to a mesh	sensor binding table [P]	Report needs_binding for the component highlight; the diagnosis itself is unaffected
Ground truth	The simulator scenario label is not available to the agent	diag.sim hidden by design [P]	Reason from channel evidence only

1.2 Quantities computed in every diagnosis

Table 2. Feature definitions used by the diagnostic procedure

window = last N cycles of samples, N >= 5, aligned to the 8 s cycle (phase p = (t mod 8)/8)	
M	= mean(window) # cycle mean (Nm)
S	= (M - 13.0) / 10.5 # severity index
P_k	= max of samples in cycle k ; Pbar = mean(P_k) # per-cycle peak
Min_k	= min of samples in cycle k ; Minbar = mean(Min_k)
A6	= 2*hypot(mean((x-M)*cos(12 pi p)), mean((x-M)*sin(12 pi p))) # 6x cycle (0.75 Hz)
sigma	= std of (x_k(p) - x_(k-1)(p)) / sqrt(2) # same-phase cycle-to-cycle noise estimate
dTCP	= mean(TOOL_CENTER_POINT_DEVIATION) - 0.104 # mm, corroboration
step	= max x_i - x_(i-1) over window # Nm per 0.5 s
idle	= mean torque while axis is holding (if available)

2. Diagnostic Workflow

The procedure has seven stages D1 to D7. Stages D1 and D2 are gates: if either fails, the agent reports the gate result instead of a mechanical diagnosis. Stage D3 computes features, D4 assigns the level, D5 and D6 generate and separate candidates, and D7 produces the report.

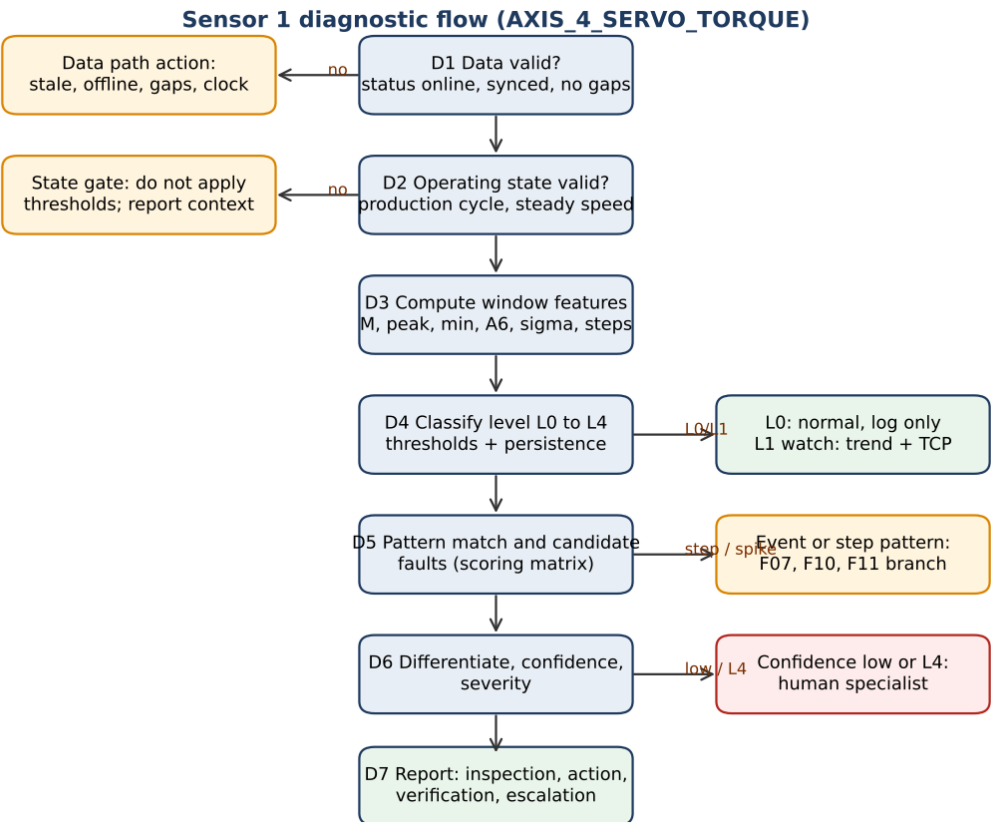


Figure 1. Diagnostic flow for Sensor 1.

Table 3. Workflow stages

Stage	Question answered	Main inputs	Output	Document reference
D1 Data validity	Can the data be trusted?	status, seq, synced, diag	Gate pass or F12 report	Document 01 Section 13
D2 Operating state	Is the robot in a state where thresholds apply?	program, speed, warm-up	Gate pass or state-context report	Document 01 Section 10
D3 Features	What happened in the window?	samples	M, S, peak, A6, sigma, step, TCP	This document Section 1
D4 Level	How abnormal is it?	features and thresholds	L0 to L4 and LS	Document 01 Sections 6, 7, 16
D5 Candidates	What could cause it?	pattern	Candidate list F01 to F12	Document 02
D6 Differentiation	Which candidate is most likely and how sure are we?	scoring matrix	Ranked diagnosis with confidence	Section 7 of this document
D7 Report	What should be done?	diagnosis, severity	Inspection, action, verification, escalation	Documents 04 and 05

2.1 Step-by-step procedure

Table 4. Detailed procedure steps

Step	Action	Decision	If the decision fails
1	Read device status, last seq, bootId, synced and diag fields for robot-weld-01	Is status online and not stale?	R-01: report offline or stale and stop
2	Count samples in the last 40 s; compare with 80 expected at 500 ms	Is at least 90 percent present?	R-02: mark incomplete
3	Check synced and the identity of the source	Is synced true and the source unique?	R-03, R-04
4	Determine the operating state from program, speed override and time since start	Production at steady speed after warm-up?	R-06: state gate
5	Check cycle periodicity of the torque trace (8 s)	Periodic?	R-07
6	Compute M, S, Pbar, Minbar, A6, sigma, step	-	-
7	Compare with baseline; compute dM, dSw, dTCP	Inside healthy tolerances?	If yes: L0, stop
8	Evaluate persistence on the peak envelope	Two consecutive cycles at or above the limit?	Else watch or transient
9	Assign level L1 to L4 or LS	-	-
10	Classify pattern P0 to P8	-	-
11	Score candidates with the evidence-weight matrix	-	-
12	Determine margin and confidence	Margin at least 2 and points at least 6?	Apply decisive test
13	Rank inspections (priority table), set severity and escalation	-	-
14	Produce report: diagnosis, evidence for and against, inspection, action, verification criteria, confidence	-	-

2.2 Prediction of time to the next threshold

When M has risen over three or more successive windows, the time to the next threshold is estimated by linear extrapolation: $t_{\text{next}} = (M_{\text{threshold}} - M_{\text{now}}) / \text{slope}$, where slope is the least-squares slope of window means. The agent should report the range obtained with the slope varied by plus and minus 30 percent. Example [D]: in the simulator ramp of 120 s the slope is 0.0875 Nm/s; at M = 15.0 Nm the first warn peak (M 18.0 Nm) is 34 s away, and the first alarm peak (M 21.6 Nm) is 75 s away. Example [E][I]: a real drift of 0.1 Nm per day from M = 15.0 Nm reaches M = 18.0 Nm after 30 days and M = 21.6 Nm after 66 days; any acceleration in the slope reduces these times and must be reported.

Table 5. Time-to-threshold formulae

Target	M threshold (Nm)	Formula	Use
Early degradation	15.0	$(15.0 - M) / \text{slope}$	Plan inspection
First warn peak	18.0	$(18.0 - M) / \text{slope}$	Schedule stop
First alarm peak	21.6	$(21.6 - M) / \text{slope}$	Stop planning
Critical	23.5	$(23.5 - M) / \text{slope}$	Escalation lead time

3. Initial Telemetry Assessment

3.1 Data validity rules

R-01: IF the device status is offline or stale THEN stop; report no current Sensor 1 value and the time of the last sample; do not diagnose.

R-02: IF more than 10 percent of the expected samples in the window are missing (seq gaps) THEN mark the window incomplete; use only M and Pbar and lower the confidence by one level.

R-03: IF synced is false for the window THEN skip A6 (phase unreliable) and use M, peak, sigma and step only.

R-04: IF two sources announce robot-weld-01 (identity conflict) THEN stop; resolve identity before diagnosis.

R-05: IF all three channels (torque, TCP, gun temperature) change implausibly at the same time (for example all flatline or all jump) THEN classify as F12 data-path fault, not as an axis fault.

Table 6. Initial assessment checks

Check	Pass condition	Fail result	Rule
Value within declared range	0 to 45 Nm	Out-of-range or clamped values flag F11	R-05, R-20
Variation present	Sigma at least 0.3 Nm and cycle swing visible	Flatline: F11 or idle state	R-19
Resolution	1 decimal	Truncated or integer values: scaling fault	R-20
Cycle periodicity	Autocorrelation peak at 8 s	No periodicity: not in production cycle or signal fault	R-07
Interval	500 ms (or announced interval)	Changed interval alters samples per cycle	R-08

4. Baseline Comparison and Threshold Evaluation

4.1 Baseline comparison

Table 7. Baseline comparison table

Feature	Baseline	Tolerance (healthy)	Deviation class	Next step
M	13.0 Nm	12.5 to 13.5	Mild 13.5 to 15.0; early 15.0 to 18.0; warning-equivalent 18.0 to 21.6; alarm-equivalent above 21.6	Compute S
Pbar	21.7 Nm	20.9 to 22.4 (5th to 95th percentile)	Above 23.5: abnormal	Compare to 26 and 30
Minbar	4.3 Nm	3.5 to 5.0	Above 5.0: baseline shift	Support M
A6	0 Nm	at most 0.2	0.4 or above: ripple present	Gear-mesh evidence
sigma	0.45 Nm	0.40 to 0.55	Above 0.55: roughness	Support
dTCP	0 mm (0.104 mm absolute)	within 0.05	Above 0.15: kinematic effect	Corroboration
step	up to 7.2 Nm per 0.5 s	at most 8	Above 8 repeating: abnormal events	Event branch

4.2 Threshold evaluation

Table 8. Threshold evaluation: condition to classification to action

Condition	Sensor value	Duration	Machine state	Diagnostic classification	Recommended action
C01	All samples 3.1 to 23.1; M 12.5 to 13.5; A6 at most 0.2	5 cycles	Production, steady	Normal (L0)	Log; no action
C02	M 13.5 to 15.0 or A6 0.4 to 0.6 or sigma above 0.55; peak below 23.7	5 cycles (40 s)	Production, steady	Watch (L1), stage 1	Increase monitoring; compare TCP; no maintenance request yet
C03	M 15.0 to 18.0; peak 23.7 to 26.0	5 cycles	Production, steady	Early degradation, stage 2	Plan inspection within days; evaluate ratios
C04	Per-cycle peak at or above 26 Nm and below 30 Nm	2 consecutive cycles (16 s)	Production, steady	Warning (L2), stage 3	Open investigation; inspection at next planned stop
C05	Per-cycle peak at or above 30 Nm and below 34 Nm	2 consecutive cycles (16 s)	Production, steady	Alarm (L3), stage 4	Recommend intervention in the shift; reduce duty; plan stop
C06	Any sample at or above 34 Nm, or M at or above 23.5 Nm, or alarm longer than 10 min	1 cycle confirmation	Any production state	Critical (L4)	Escalate immediately; stop at a safe point; specialist
C07	Peak at or above 26 Nm	1 cycle only, next cycle normal	Production	Transient or isolated event	Log; verify next cycles; check event log
C08	Any value 26 Nm or above	any	Not production cycle (jog, teach, warm-up)	Not classifiable	Report context; do not apply thresholds
C09	Flatline (sigma below 0.1) or values 0 or 45 at all samples	3 cycles	Production	Sensor suspect (LS)	Check data path and signal; see Document 04
C10	Values 0 to 3 Nm for whole window with program stopped	any	Idle or servo off	Not a fault	None
C11	Mean below 12.0 Nm for 5 cycles	5 cycles	Production	Low side: payload decrease, program change, offset	Check program and payload; no wear conclusion

4.3 Persistence evaluation

Persistence is evaluated on the per-cycle peak envelope because the raw signal crosses a limit for only part of each cycle (Document 01 Section 9.2). The envelope sequence P_1, P_2, ... is formed from the maximum sample in each 8 s cycle. A level opens when the envelope has been at or above the limit for 2 consecutive cycles, and clears when it has been below limit minus 1 Nm for 2 consecutive cycles.

Table 9. Persistence parameters

Level	Open	Clear	Cooldown	Max time before escalation to next level [I]
L1 Watch	5 cycles of watch features	5 cycles in healthy band	10 min	7 days (real) or 10 min (simulator) without improvement escalates to L2 review
L2 Warning	Peak at or above 26 Nm for 2 cycles	Peak below 25 Nm for 2 cycles	10 min per machine	Escalate to L3 if M rises by 3 Nm or more or peak reaches 30 Nm
L3 Alarm	Peak at or above 30 Nm for 2 cycles	Peak below 29 Nm for 2 cycles	10 min	Escalate to L4 if alarm persists 10 minutes or a sample reaches 34 Nm
L4 Critical	Sample at or above 34 Nm or M at or above 23.5 Nm	Only after inspection and verification	None	-

R-09: IF the per-cycle peak is at or above 26 Nm for 2 consecutive cycles AND the robot is in production at steady speed THEN classify as WARNING, abnormal operation, and evaluate fault candidates (Section 6).

R-10: IF the per-cycle peak is at or above 30 Nm for 2 consecutive cycles THEN classify as ALARM and recommend intervention.

R-11: IF any sample is at or above 34 Nm THEN confirm in the next cycle; IF confirmed THEN classify as CRITICAL and escalate; IF not confirmed THEN classify as a transient (C07) and recheck after 5 cycles.

5. Rate-of-Change and Operating-State Evaluation

5.1 Rate-of-change evaluation

Table 10. Rate-of-change decision table

Observation	Threshold	Interpretation	Candidates favoured	Rule
Mean M changes within one cycle	1 Nm or more in 8 s	Step change	F07, F10, F03 onset, F08, F11	R-12
M rises 0.05 to 1 Nm per minute	Fast ramp	Acute degradation or simulator injection	F01 (simulator), F02 step, F07 follow-on	R-13
M rises 0.02 to 0.5 Nm per day	Gradual drift	Progressive wear	F01 real wear, F02, F04	R-14
M follows temperature and reverses	Reversible	Thermal	F09, F02 warm-up part	R-15
Single sample step above 8 Nm and return	Within 1 s	Transient	F11 glitch, F12, F07 collision	R-16
Steps above 8 Nm at the same phase every cycle	3 cycles	Recurring event	F05, F06	R-17
A6 rises with dM	Ratio 0.2 to 0.4	Locked growth	F01	R-18

R-12: IF M changes by 1 Nm or more within one cycle AND does not continue to rise in the following 5 cycles THEN classify as a step change; check program, payload and collision events before wear (F07, F10) and exclude gearbox wear as the primary cause.

R-13: IF M rises continuously over 3 or more successive 5-cycle windows at 0.05 Nm per minute or faster AND A6 rises proportionally THEN classify as a developing GEARBOX_WEAR ramp (F01) and estimate time to the next threshold by linear extrapolation of M.

5.2 Operating-state gate

R-06: IF the robot is not in production cycle at steady speed (jog, teach, warm-up, reduced speed override, idle, servo off) THEN do not apply thresholds and do not diagnose wear; report which state prevents classification.

R-07: IF the autocorrelation of the window shows no 8 s periodicity AND the program is running THEN suspect F10 (program change) or F11 (signal fault).

R-08: IF the publish interval has been changed THEN recompute samples per cycle ($N = 8 \text{ s} / \text{interval}$); IF the interval is 5 s or more THEN A6 cannot be evaluated; use M and peak only.

Table 11. State gating table

State	Allowed features	Not allowed	Use of result
Production steady	All	None	Full diagnosis
Warm-up (first minutes)	Trend only	Absolute thresholds, baseline	Exclude from baseline [E]
Reduced speed override	Shape features scaled by speed	Peak thresholds	Report context [E]
Idle, servo enabled	Idle torque level	Cycle features	Detect idle offset (F03, F08) [E]
Jog or teach	None	All thresholds	Ignore
Servo off or brake engaged	None	All	Ignore
Gateway offline or stale	None	All	Data-path report

6. Fault Candidate Generation

6.1 Pattern classes

Table 12. Pattern classes derived from features

Class	Definition	Candidate faults (Document 02)
P1 Wear pattern	dM at least 1 Nm AND A6 at least 0.4 AND sigma at least 0.55 AND dTCP positive	F01, F05, F04
P2 Friction pattern	dM at least 1 Nm AND A6 below 0.4 AND swing unchanged	F02, F03, F09, F08, F06
P3 Idle-offset pattern	dI at least 0.5 Nm	F03, F08, F11
P4 Phase-bump pattern	Residual to template concentrated in one phase window below 1.5 s	F06, F05, F07
P5 Spike pattern	Isolated samples at least 3 Nm beyond template	F05, F04, F07, F11
P6 Step pattern	dM at least 1 Nm within one cycle	F07, F10, F03, F08
P7 Load pattern	M and swing change proportionally	F07 payload, F10
P8 Dead-signal pattern	sigma below 0.1 or constant, or all channels affected	F11, F12
P0 Healthy	All features in healthy band	None

6.2 Evidence weights

Each candidate receives points from the evidence below. The weights are proposals [I] chosen so that the project fault F01 reaches the highest score only when ripple, mean shift, noise and TCP agree.

Table 13. Evidence-weight matrix (points added to each fault when the evidence is present) [I]

Evidence	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11
dM at least 1 Nm	2	2	2	1	0	1	2	1	1	1	0
A6 at least 0.4 Nm and A6 / dM in 0.2 to 0.4	4	0	0	1	1	0	0	0	0	0	0
Noise ratio consistent with $1 + 0.143 \text{ dM}$	2	1	0	2	1	0	0	1	0	0	0
dTCP / dM in 0.03 to 0.08 mm per Nm	2	0	0	1	2	0	1	0	0	0	0
dTCP / dM above 0.15 mm per Nm (TCP rise large for the torque shift)	0	0	0	0	3	0	1	0	0	0	0
Swing unchanged	1	1	1	1	1	1	0	1	1	0	0
Swing changed	0	0	0	0	0	0	3	0	0	3	0
Idle offset at least 0.5 Nm	0	1	3	1	0	0	0	3	1	1	2
Brake or motor housing locally hot	0	1	3	1	0	0	0	0	1	0	0
Narrow spikes locked to the reversal phases (about 2.4 s and 6.0 s)	0	0	0	1	4	0	1	0	0	0	0
Smooth bump in one phase window, rest unchanged	0	0	0	0	1	4	1	0	0	1	0
Isolated spike, not repeated	0	0	0	1	0	0	3	0	0	0	2
Gun temperature unchanged	1	1	1	1	1	1	1	1	0	0	1
Onset gradual (ramp or drift)	2	1	0	1	1	1	0	0	1	0	0
Onset step (within one cycle)	0	1	2	0	1	1	3	2	0	3	3
Reversible with temperature	0	1	0	0	0	0	0	0	3	0	0
Flatline or clipping	0	0	0	0	0	0	0	0	0	0	4
Other channels affected too	0	0	0	0	0	0	1	0	1	1	0
Event log shows collision or safety stop	0	0	0	0	0	0	5	0	0	0	0
Event log shows planned program or tool change	0	0	0	0	0	0	2	0	0	5	0

R-19: IF sigma is below 0.1 Nm for 3 cycles AND the program is running THEN add 4 points to F11 and classify as sensor suspect.

R-20: IF values are clamped at 0 or at 45 Nm for 3 cycles THEN classify as F11 saturation, not as a torque event.

7. Fault Differentiation

7.1 Differentiation procedure

- Sum the points for each candidate (Table of evidence weights). The candidate with the highest total is the leading diagnosis, the next highest is the alternative.
- Compute the margin between the first and second candidate. A margin of 4 points or more is clear, 2 to 3 is ambiguous, 0 to 1 is unresolved.
- For an ambiguous or unresolved result, apply the decisive test for the pair in Document 02 (ambiguity table) and recommend the corresponding inspection first.
- Always state the evidence for the leading candidate and the evidence against the leading alternative.

Table 14. Fault differentiation matrix: decisive observations

Question	Answer favouring F01	Answer favouring alternatives
Is there a 0.75 Hz ripple?	Yes, 0.2 to 0.4 Nm per Nm of dM	No: F02, F03, F09, F10, F07 payload
Is the swing unchanged?	Yes	No: F07, F10
Does TCP rise with dM at 0.03 to 0.08 mm per Nm?	Yes	No: F02, F03, F08, F09; much higher: F05
Is the idle torque raised?	No (F01 not modeled at idle)	Yes: F03, F08, F11
Does M follow temperature reversibly?	No	Yes: F09, F02 warm-up part
Is the onset a step?	No, ramp	Yes: F07, F10, F03, F08, F11
Are spikes locked to reversal phases?	No	Yes: F05
Is the pattern identical on all channels?	No	Yes: F12

7.2 Decision matrix: pattern by level

Table 15. Diagnostic decision matrix

Pattern	Watch (L1)	Warning (L2)	Alarm (L3)	Critical (L4)
P1 wear pattern	Trend M and A6 every shift; compare TCP	Inspect gearbox at next stop (oil sample, backlash)	Intervene in shift; reduce duty; plan gearbox replacement	Stop; specialist; do not restart before inspection
P2 friction pattern	Check lubrication records	Inspect lubricant, seals, temperature	Relubricate or repair in shift	Stop if temperature rises; specialist
P3 idle offset	Check brake and drive offset at next stop	Inspect brake and current sensors	Repair in shift	Stop; check brake heat
P4 phase bump	Inspect dress pack	Inspect dress pack and backlash	Inspect immediately	Stop if cable damage
P5 spikes	Log and watch	Inspect fasteners and coupling	Inspect in shift	Stop; mechanical damage possible
P6 step	Check program change	Check payload and collision	Inspect after collision	Stop; inspect
P7 load	Re-baseline if planned change	Verify tool data	Verify payload	Stop if overload
P8 dead signal	Check data path	Check gateway	Check gateway and signal	Escalate; blind spot

8. Inspection Prioritisation, Severity and Escalation

8.1 Inspection priority

Table 16. Inspection priority table

Prior ity	Leading diagnosis	First inspection	Why first	Time needed [I]	Stop required
1	F07 after suspected collision	Visual check for damage, tool data	Safety and damage	10 to 20 min	Yes
2	F01 gearbox wear	Oil sample and magnetic plug; backlash	Confirms or excludes particles cheaply	30 to 60 min	Yes, planned stop
3	F03 brake drag	Brake release voltage	Quick electrical check	15 min	Yes
4	F02 lubrication	Lubricant level and condition	Low cost	20 min	Yes
5	F05 looseness	Fastener torque, backlash	Cheap and decisive	30 min	Yes
6	F06 external drag	Dress pack	Visual	10 min	Short stop
7	F08 drive fault	Drive diagnostic log	No mechanical opening	15 min	No
8	F04 bearing	Play, temperature	Needs tools	45 min	Yes
9	F11 / F12	Signal path and gateway diag	Data first	10 min	No

8.2 Severity assessment

Table 17. Severity score [I]

Factor	Score 0	Score 1	Score 2	Score 3
Platform level	L0	L1	L2	L3 or L4
Severity index S	below 0.05	0.05 to 0.48	0.48 to 0.82	above 0.82
Trend	stable	slow drift	fast ramp	step or accelerating
TCP	healthy	above 0.15 mm	above 0.25 mm	above 0.5 mm
Consequence	none	quality risk	downtime risk	safety or damage risk

Severity total 0 to 3: low; 4 to 7: medium; 8 to 11: high; 12 to 15: critical [I]. A critical level L4 overrides the sum.

8.3 Escalation criteria

Table 18. Escalation matrix

Trigger	Severity	Escalate to	Time target [I]	Action
L1 watch persists 7 days	Low	Maintenance planner	next planning cycle	Schedule inspection
L2 warning	Medium	Maintenance technician	next planned stop	Inspection request
L3 alarm	High	Maintenance supervisor	within shift	Intervention; reduced duty
L4 critical or collision	Critical	Specialist and safety officer	immediately	Stop at safe point; lock-out; inspect
Sensor suspect LS	Medium	Controls engineer	within shift	Data-path check
Unresolved diagnosis (margin 0 to 1)	Medium	Specialist	next stop	Decisive test
Alarm not cleared after repair	High	Specialist	immediately	Recheck diagnosis

9. Corrective Action, Verification and Confidence

9.1 Recommended corrective action

Table 19. Diagnosis to recommended action

Diagnosis	Recommended action	Constraint	Document 04 section
F01 gearbox wear	Inspect at next stop; replace gear set or gearbox if confirmed; do not run to alarm	OEM procedure, lock-out and tag-out	Corrective actions
F02 lubrication	Relubricate or replace lubricant, seals	OEM grade	Lubrication
F03 brake drag	Repair brake	OEM	Electrical inspection
F04 bearing	Replace bearings	OEM	Mechanical inspection
F05 looseness	Retorque, replace coupling	Torque spec	Alignment and installation
F06 external drag	Re-dress, clean	None	Visual inspection
F07 overload or collision	Inspect, re-master	Safety	Escalation
F08 drive fault	Replace sensor or drive	OEM	Electrical inspection
F09 thermal	Cooling, warm-up	None	Troubleshooting
F10 program change	Re-baseline	None	Calibration
F11 sensor fault	Fix signal	None	Sensor inspection
F12 data fault	Fix network, clock, identity	None	Troubleshooting data path

9.2 Post-maintenance verification criteria

Table 20. Post-maintenance verification criteria for Sensor 1

Criterion	Pass condition	Window	Fail action
Cycle mean	12.5 to 13.5 Nm	20 cycles (160 s)	Repeat inspection
Per-cycle peak	at most 23.1 Nm	20 cycles	Investigate residual friction
Cycle minimum	3.5 to 5.0 Nm	20 cycles	Investigate offset
A6	at most 0.2 Nm	5 cycles	Gear mesh still suspect
sigma	0.40 to 0.55 Nm	20 cycles	Roughness remains
Platform status	normal for 600 s with no warn flicker	600 s (matches the simulator test)	Escalate
TCP	at most 0.15 mm mean	20 cycles	Mastering or backlash check
Gun temperature	37 to 58 degC after warm-up	10 min	Process check
Repeat after production	Same criteria after 24 h	24 h	Re-open
Simulator	After NORMAL command the ack shows scenario NORMAL and mean returns to 13.0 within one cycle	1 cycle	Send reboot command

9.3 Diagnostic confidence

Table 21. Confidence scoring [I]

Total points for leading candidate	Margin to second candidate	Data quality	Confidence	Report wording
9 or more	4 or more	complete	High	Diagnosis stated with evidence
6 to 8	2 or more	complete	Medium	Probable diagnosis; recommend confirming inspection
3 to 5	any	complete	Low	Possible; list alternatives
below 3	any	any	Insufficient	State that the evidence is insufficient
any	any	incomplete window or synced false	Reduce by one level	State data limitation

9.4 Conditions requiring human or specialist intervention

- Any L4 critical condition; any suspected collision; any alarm persisting more than 10 minutes.
- Unresolved diagnosis (margin 0 to 1) or low confidence at L3.
- Any condition in which the sensor itself is suspect and the machine is in alarm.
- Any proposed action that requires opening the drive, removing guards or working inside the cell: lock-out and tag-out and the OEM procedure apply; the agent cannot act on the machine and only issues reports and user-interface commands [P].

10. Worked Scoring Examples

The four examples apply the evidence-weight matrix to representative windows. They show the arithmetic the agent should perform and what to report. Points are listed per candidate in the order F01, F02, F03, F04, F05, F06, F07, F08, F09, F10, F11.

Table 22. Example A: wear pattern (M 18.2 Nm, A6 1.5 Nm, sigma 0.79 Nm, TCP mean 0.376 mm, swing unchanged, gun temperature unchanged, ramp onset)

Item	Value	Reading
dM	+5.2 Nm	at least 1 Nm: evidence present
A6 / dM	0.29	inside 0.2 to 0.4: evidence present
kN	1.76 (expected $1 + 0.143 \times 5.2 = 1.74$)	consistent
dTCP / dM	$0.272 / 5.2 = 0.052$ mm per Nm	inside 0.03 to 0.08
Level	peak about 26.2 Nm in half of the cycles	L2 warning
Scores F01 to F11	14, 6, 4, 8, 7, 4, 4, 4, 3, 1, 1	F01 leads by 6 points over F04
Confidence	High (14 points, margin 6)	Diagnosis: GEARBOX_WEAR developing, stage 3
Action	Inspect gearbox at next stop (oil sample, backlash); trend M every shift	Verification per Section 9.2

Table 23. Example B: friction pattern without ripple (M 16.0 Nm, A6 0.15 Nm, sigma 0.55 Nm, TCP unchanged, idle offset +1.0 Nm, M higher after cold start)

Item	Value	Reading
dM	+3.0 Nm	evidence present
A6	0.15 Nm	below 0.4: no ripple
kN	1.22 (F01 would give 1.43)	inconsistent with F01 ratio
dTCP / dM	0.003 mm per Nm	below 0.03: absent
Level	peak about 23.5 Nm	stage 2, platform status normal
Scores F01 to F11	6, 7, 7, 5, 3, 4, 3, 6, 7, 2, 3	F02, F03 and F09 tie at 7; margin 0
Confidence	Low; unresolved	Friction family, F01 not supported
Action	Apply decisive tests: warm-up comparison (F09 vs F02), brake supply voltage and release timing (F03), lubricant condition (F02); inspect lubricant first	Re-score after tests

Table 24. Example C: step change (M jumps from 13.0 to 19.0 Nm in one cycle, swing +2.1 Nm, A6 0.1 Nm, gun temperature also changed; event log not yet consulted)

Item	Value	Reading
dM	+6.0 Nm in 8 s	step
Swing	+2.1 Nm (30 percent)	changed
Other channels	also changed	yes
Scores F01 to F11	2, 3, 4, 1, 1, 2, 9, 3, 2, 8, 3	F07 9, F10 8; margin 1
Confidence	Low; unresolved	Context decides. If the event log then shows a planned program or tool change, F10 gains 5 and F07 gains 2 (F10 13 versus F07 11, margin 2, ambiguous: confirm tool data); a collision entry gives F07 14 versus F10 8
Action	Check program change log and tool or payload data first; if a planned change, classify as F10 and re-baseline; if not planned, inspect for collision (F07)	No wear conclusion from this window

Table 25. Example D: flat signal (13.0 Nm constant for 5 cycles, sigma 0.0, program running, other channels normal)

Item	Value	Reading
sigma	0.0 Nm	below 0.1 Nm: rule R-19 applies
Scores F01 to F11 before R-19	0, 0, 0, 0, 0, 3 (step), 2, 0, 3, 3 (step) + 1 (gun unchanged)	F11 4 after step and gun; R-19 adds 4, so F11 total 8
Level	LS sensor suspect	Not a mechanical level
Action	Check torque data path, gateway diag fields and controller state before any mechanical work	Document 04 data-path table

11. Fault-Specific Diagnostic Checklists

Table 26. Checklist: confirm, refute and decisive test per fault

Fault	Confirm when all hold	Refute when any holds	Decisive test
F01 gearbox wear	dM at least 1 Nm; A6 at least 0.4 and A6/dM 0.2 to 0.4; kN near 1 + 0.143 dM; dTCP positive; swing unchanged; gun unchanged; ramp onset	Idle offset large; swing changed; M reversible with temperature; A6 below 0.2 with dM above 3	Oil particles, backlash
F02 lubrication	dM at least 1; A6 below 0.4; larger when cold; idle offset small	A6/dM above 0.2; metal in oil	Lubricant condition and warm/cold comparison
F03 brake drag	Idle offset at least 1 Nm; mean up; A6 absent; brake warm	Idle offset absent; brake cool	Brake voltage and release timing
F04 bearing	Noise rise with sparse spikes; A6 low; dM small	Clear 0.75 Hz line; no spikes	Bearing play, temperature
F05 looseness	Spikes at the same 1 or 2 phases; dTCP high relative to dM	Mean shift above 3 Nm with ripple	Fastener torque, backlash
F06 external drag	Bump in one phase window; rest unchanged	Whole-cycle shift	Dress pack check
F07 overload/collision	Step or spike; swing changed (payload) or TCP step (collision)	Gradual ramp with ripple	Visual, tool data, event log
F08 drive fault	Offset at idle; no TCP change; non-harmonic lines	Brake warm; TCP changed	Drive diagnostics
F09 thermal	M follows temperature; reversible	Irreversible shift	Temperature trend
F10 program change	Step at program boundary; shape change	No program change	Program log
F11 sensor fault	Flat, clipped or burst; other channels normal	Periodic cycle present	Raw gateway log
F12 data path	Gaps, stale, synced false, all channels	Complete data	Gateway diag

12. Reporting, Concurrent Faults and Alert Classification

12.1 Alert classification table

Table 27. Alert classification

Level	Name	Trigger	Twin appearance (project behaviour)	Agent behaviour	Response
L0	Normal	Features healthy	Normal tint	Log	None
L1	Watch	M 13.5 to 15.0 or A6 0.4 to 0.6	Normal tint (no platform state)	Note in report; no alert	Trend
L2	Warning	Peak at least 26 for 2 cycles	Amber tint #d98400 steady; triangle marker	Open investigation	Inspect at next stop
L3	Alarm	Peak at least 30 for 2 cycles	Red tint #b3261e pulsing; square marker; status message	Investigation with intervention	Intervene in shift
L4	Critical	Sample at least 34, M at least 23.5, alarm over 10 min	Same as alarm; agent highlight planned	Escalate	Immediate
LS	Sensor suspect	Flat, clipped, or noisy signal	Marker unchanged	Data-path report	Check path

12.2 Concurrent faults and conflicting evidence

- Superposition: when a step change and a ramp occur together, explain and subtract the step first (F10 or F07), then re-evaluate the residual drift for A6 and noise.
- Conflicting channels: if TCP is high but Sensor 1 is normal, evaluate F05 (looseness) and mastering errors before wear; if Sensor 1 is high but TCP and A6 are normal, favour F02, F03, F09.
- A fault can mask another: an offset fault (F08, F11) shifts M and can make wear appear more severe; check idle offset before computing S.
- Never combine the severity of a data fault with a mechanical fault into one score; report them separately.

12.3 Questions the agent may ask when context is missing

Table 28. Clarifying questions

Missing information	Question	Why it matters
Operating state	Is the robot in production at full speed and warmed up?	Gates the thresholds
Program	Was the program, payload or tool changed in the last shift?	Distinguishes F10 and F07 from wear
Events	Was there a collision, emergency stop or brake fault?	F07, F03
Maintenance	When was axis 4 last lubricated and what grade?	F02
Temperature	What are the gearbox and ambient temperatures?	F09
Idle	What is the holding torque at standstill?	F03, F08

12.4 Agent report template

Table 29. Example structured diagnostic report

component_tag	: robot_forearm_axis_4 (resolved via ROBOT-WELD-01.AXIS_4_SERVO_TORQUE; needs_binding if unbound)
level	: L2 warning severity: medium
window	: 40 s (5 cycles), synced true, 0 percent missing
features	: M 18.2 Nm (S 0.50), Pbar 26.1 Nm, A6 1.5 Nm, sigma 0.79 Nm, dTCP +0.27 mm
diagnosis	: GEARBOX_WEAR (F01) developing, stage 3 confidence: High (14 points, margin 6)
evidence for	: mean shift, 0.75 Hz ripple in ratio 0.29, noise ratio 1.76, TCP rise, swing unchanged, gun temperature un
evidence vs	: none material; F04 bearing is second (8 points) but has no spikes
inspection	: gearbox oil sample and magnetic plug, backlash at flange (Document 04)
action	: inspect at next planned stop; do not run to alarm; OEM procedure and lock-out apply
verification	: M 12.5 to 13.5 Nm, A6 at most 0.2 Nm, peak at most 23.1 Nm, status normal for 600 s
citations	: Document 01 Section 7; Document 02 F01; Document 03 R-21

12.5 Complete decision procedure in pseudocode

Table 30. Pseudocode of the Sensor 1 diagnostic procedure

```
def diagnose(window, ctx):
    if ctx.status in (offline, stale) or ctx.identity_conflict: return report_data_path()
    if missing_fraction(window) > 0.10: flag_incomplete()
    if not ctx.synced: skip(A6)
    if ctx.state != production_steady: return report_state_gate(ctx.state)
    f = features(window)
    level = classify_level(f)
    if level == L0: return log_normal()
    pattern = classify_pattern(f)
    score = {fault: sum(weights[e][fault] for e in evidence(f)) for fault in F01..F11}
    if f.sigma < 0.1: score[F11] += 4
    top, second = ranked(score)[:2]
    margin = score[top] - score[second]
    conf = confidence(score[top], margin, data_quality(window))
    plan = inspections(top, second, margin)
    return report(level, pattern, top, second, conf, plan, verification_criteria(top))
```

13. Edge Cases, False-Positive Control and Verification Runs

13.1 Window length, false positives and detection power

A shorter window reacts faster but produces more false watch indications. The values below come from replicating the Sensor 1 equations with noise (300 seeded windows per case) [D]; they justify the 5-cycle default window for feature evaluation.

Table 31. Window length trade-off for the watch features [D]

Window	M standard deviation (healthy)	A6 healthy mean / 95th percentile	A6 at S 0.1 mean	Chance a healthy window shows A6 at least 0.4 Nm	Use
1 cycle (8 s, 16 samples)	0.11 Nm	0.20 / 0.39 Nm	0.35 Nm	about 5 percent	Preliminary only; flag as low confidence
5 cycles (40 s, 80 samples)	0.05 Nm	0.09 / 0.18 Nm	0.30 Nm	well below 1 percent	Default feature window
10 cycles (80 s, 160 samples)	0.04 Nm	0.07 / 0.12 Nm	0.30 Nm	negligible	Trend confirmation; real-robot baseline

13.2 Edge cases

Table 32. Edge cases and how the procedure handles them

Situation	Risk of misdiagnosis	Handling
Window straddles a fault injection or a program change	Mixed statistics, false ripple	Split the window at the change point; evaluate each part
Gateway reboot inside window (new bootId)	Phase discontinuity	Restart window after reboot; do not compute A6 across the boundary
Clock not synced at start	Receive-time stamps jitter	Wait for synced true; use M, peak, sigma
Publish interval changed to 250 ms or 5000 ms	Wrong samples per cycle	Recompute $N = 8 \text{ s} / \text{interval}$; skip A6 at 5000 ms
Per-sample status flickers between normal and warn	Operator assumes noise	Use per-cycle envelope; if M is above 15 Nm treat as persistent warning
Peak at 26.1 Nm but M in healthy band	Isolated outlier mistaken for wear	Require 2 consecutive cycles and M above 13.5
TCP high, torque normal	Wear or mastering error assumed	Check dTCP/dM; evaluate F05 and mastering first
Torque high, TCP normal and A6 absent	Gearbox wear assumed	Evaluate F02, F03, F09, F08
Gun temperature rising while torque normal	Wrong channel blamed	Gun temperature is not affected by GEARBOX_WEAR; evaluate the gun channel separately
Simulator and board both running	Identity conflict	Stop one source
Binding absent on the twin	No highlight	Report needs_binding; diagnosis continues
Retained birth message limits differ from this document	Wrong thresholds	Use the live birth limits for platform status and keep this document's feature thresholds

13.3 Corroboration scenarios

Table 33. Torque level against corroborating channels

Sensor 1 level	TCP rising, gun normal	TCP normal, gun normal	Gun abnormal (TCP any)
Watch (L1)	Early F01 likely; keep trending	F02, F03, F09, F10 more likely; check idle	Process or thermal event; evaluate gun first
Warning (L2)	F01 likely (high confidence if A6 present)	Friction family; inspect lubricant	Mixed; separate the channels
Alarm (L3)	F01 stage 4; intervene	Friction or overload; inspect urgently	Escalate; possible process fault
Critical (L4)	F01 beyond model or F07; stop	F07 collision or sensor saturation; stop	Stop; specialist

13.4 Verification run procedure

Table 34. Verification run after maintenance

Step	Action	Pass criterion
1	Remove lock-out per site procedure; restore the robot to production mode at 100 percent speed	Robot in production cycle
2	Warm up for at least 3 cycles (24 s in the simulator; 15 to 30 minutes on a real robot [E])	Warm-up complete
3	Collect 20 cycles (160 s) of Sensor 1, TCP and gun temperature	Window complete, synced true
4	Compute features and compare with the post-maintenance criteria	All criteria met
5	Run 600 s and confirm no warn or alarm samples	Status normal
6	Recheck after 24 h of production	Criteria met again
7	Update baseline record and close the investigation	Baseline stored

14. Response Playbooks by Level

Each playbook lists what the agent reports, what it asks the maintenance organisation to do, and what data it must collect so that a later investigation can reuse the evidence.

Table 35. Playbook L1 Watch

Item	Content
Trigger	M 13.5 to 15.0 Nm, or A6 0.4 to 0.6 Nm, or sigma above 0.55 Nm, for 5 cycles, no platform warn
Immediate agent actions	Log observation with feature values; fetch TCP and gun temperature; compute S; store window for baseline comparison
Report content	Stage 1 incipient; candidate list ranked; statement that no limit is crossed; time-to-threshold estimate if three windows are available
Maintenance request	None yet; add to the next planned inspection list; ask whether lubrication is due
Data to preserve	Raw samples of the window, TCP mean, state context
Re-evaluation	Every 5 cycles in the simulator; every shift on a real robot [I]
Escalation condition	M increases by 1 Nm or more between consecutive windows, or A6 exceeds 0.6 Nm, or the watch persists for 7 days [I]

Table 36. Playbook L2 Warning

Item	Content
Trigger	Per-cycle peak at or above 26 Nm and below 30 Nm for 2 consecutive cycles
Immediate agent actions	Open one investigation (10 min cooldown); compute candidate scores; check TCP, idle, temperature
Report content	Level, stage 3 or earlier, diagnosis with confidence, evidence for and against, time-to-alarm estimate
Maintenance request	Inspection at the next planned stop; list the first inspection from the priority table
Operational advice	Continue production only if the stop is planned within the estimated time to alarm; avoid increasing speed or payload
Data to preserve	Window features, ratios, TCP trend
Escalation condition	Peak at or above 30 Nm, M rise of 3 Nm or more since the investigation opened, or confidence Low with F07 candidate

Table 37. Playbook L3 Alarm

Item	Content
Trigger	Per-cycle peak at or above 30 Nm and below 34 Nm for 2 consecutive cycles
Immediate agent actions	Raise the alarm in the investigation; confirm data validity; compute severity score
Report content	Stage 4; diagnosis; recommend intervention in the shift; reduced duty; planning of the stop and of spare parts (gear set, lubricant, seals)
Maintenance request	Intervention within the shift; lock-out and OEM procedure
Operational advice	Reduce speed override or duty if the process allows; do not run to failure
Data to preserve	Full window, TCP trend, alarm time, event log
Escalation condition	Sample at or above 34 Nm, alarm for 10 minutes, or noise or temperature rising rapidly

Table 38. Playbook L4 Critical

Item	Content
Trigger	Any confirmed sample at or above 34 Nm, or M at or above 23.5 Nm, or alarm longer than 10 minutes, or suspected collision
Immediate agent actions	Escalate to specialist and safety officer; mark investigation critical
Report content	Possible damage; stop at a safe point; do not restart before inspection
Maintenance request	Immediate lock-out; full mechanical inspection
Data to preserve	Everything; avoid overwriting history (telemetry TTL 48 h in the platform [P], so export evidence)
Escalation condition	Not applicable; restart only after verification criteria are met

Table 39. Playbook LS Sensor suspect

Item	Content
Trigger	Flat, clipped or burst signal; or all channels affected together
Immediate agent actions	Check gateway diag fields (uptime, rssi, heapFree, reconnect counters), seq, status, identity
Report content	Data-path report; no mechanical conclusion
Maintenance request	Controls or IT check of the path; Document 04 data-path table
Operational advice	Machine continues only if other channels and procedures cover the risk
Escalation condition	Alarm on other channels or no data for more than 10 minutes

15. Reasoning Chains for Retrieval

The chains below state complete diagnostic reasoning in sentence form so that a retrieval system can return them as supporting evidence.

Table 40. Reasoning chains

Observation	Reasoning	Conclusion
M 17 Nm, A6 1.2 Nm, TCP 0.33 mm, gun unchanged, swing 7.0 Nm	The mean rise of 4 Nm gives S 0.38; expected A6 is $3.0 \times 0.38 = 1.14$ Nm and expected TCP mean is $0.104 + 0.55 \times 0.38 = 0.31$ mm; both match, the swing is unchanged and the gun is unaffected	Gearbox wear (F01) at stage 2, no platform limit yet; inspect gearbox at a planned stop
M 17 Nm, A6 0.1 Nm, TCP 0.11 mm	Mean rise without the expected ripple (1.14 Nm) or TCP rise (0.31 mm) is inconsistent with the F01 ratios	Friction family (F02, F03, F09); not F01
Peak 27 Nm in one cycle only, M 13.1 Nm	A single peak with a healthy mean cannot be a baseline shift; healthy peaks reach at most 23.1 Nm so it is an outlier	Transient; recheck 5 cycles; check data path
Peak 27 Nm every cycle, M 19.4 Nm	M of 19.4 Nm is S 0.61 and the peak of 27.2 Nm matches the model at S 0.6 (27.36 Nm)	Warning stage 3; investigate for F01
Platform shows warn/normal flicker once per cycle	Per-sample classification flickers because only the peak crosses 26 Nm; this occurs at S 0.4 to 0.6	Persistent early warning, not noise
Torque trace flat at 13.0 Nm, sigma 0	A healthy axis shows sigma 0.45 Nm and a swing of 7 Nm; zero variance means the value is not following the axis	Sensor or data fault (F11), not a healthy axis
Torque 0 to 2 Nm for hours, program stopped, status online	No cycle and a low holding torque	Idle, not a fault
M +6 Nm and swing +30 percent after a program change	Mean and swing change together and no ripple	F10 or F07 payload; re-baseline after confirming
Spikes of 5 Nm at 2.4 s and 6.0 s every cycle, M 13.3 Nm	Spikes lie at the torque reversal phases; mean unchanged	Backlash or looseness (F05)
Bump of 3 Nm at 1.2 s only	One phase window, rest unchanged	External drag (F06)
M 16 Nm after cold start, falls to 14 Nm in 20 minutes	Reversible friction component	Thermal or lubrication (F09, F02 warm-up part)
Sample at 38 Nm followed by baseline, TCP step +0.3 mm	Single impulsive event with kinematic step	Collision (F07); stop and inspect
Torque, TCP and gun all constant, seq gaps	All channels affected at once	Data path (F12)
Alarm after repair with M 13.0 Nm and A6 0.1 Nm	Features healthy but alert not cleared	Alert clearing latency or stale state; verify 2-cycle clear rule and status

16. IF/THEN Rule Library

Table 41. Consolidated IF/THEN rule library

Rule	IF	THEN
R-01	status offline or stale	stop; report data-path state
R-02	more than 10 percent samples missing	mark incomplete; lower confidence
R-06	state not production steady	do not apply thresholds
R-09	peak at least 26 Nm for 2 cycles	WARNING; evaluate candidates
R-10	peak at least 30 Nm for 2 cycles	ALARM; intervene
R-11	sample at least 34 Nm confirmed	CRITICAL; escalate
R-12	M step of 1 Nm or more in one cycle	check program, payload, collision first
R-13	M ramp with proportional A6	F01 developing; extrapolate time to threshold
R-21	A6 at least 0.4 AND A6/dM in 0.2 to 0.4 AND dTCP positive AND swing unchanged AND gun temperature unchanged	leading diagnosis F01, high confidence if points at least 9
R-22	dM at least 1 Nm AND A6 below 0.2 AND swing unchanged AND M follows temperature	F09 or F02 warm-up component
R-23	dM at least 1 Nm AND A6 below 0.2 AND idle offset at least 0.5 Nm	F03 brake drag or F08 drive offset
R-24	spikes at least 3 Nm at the same 1 or 2 phases for 3 cycles AND dM below 1 Nm	F05 backlash or looseness
R-25	bump at one phase window AND the rest of the cycle unchanged	F06 external drag
R-26	mean and swing both change proportionally AND A6 below 0.2	F07 payload or F10 program
R-27	one sample above 30 Nm followed by baseline AND TCP step	F07 collision; escalate
R-28	dM at least 1 Nm AND TCP unchanged AND A6 below 0.2 AND idle offset	F08 or F11 offset
R-29	TCP mean above 0.25 mm AND torque not yet in warn	early wear signal; open L1 or L2 watch on F01
R-30	TCP above 0.5 mm AND dM above 8.6 Nm	alarm-equivalent F01
R-31	status flickers warn/normal at cycle rate AND M above 15 Nm	not noise; treat as L2 persistent warning
R-32	only the warn flag is on while M is in the healthy band	check for glitch or per-sample outlier; no wear
R-33	component not bound on the twin	report needs_binding, no highlight
R-34	repair done AND verification criteria not met	reopen; do not clear alert
R-35	diagnosis ambiguous (margin 0 to 1)	apply decisive test and recommend that inspection first

17. Summary, Safety Notes, References and Revision

17.1 Diagnostic summary

- Sequence: validate data, validate state, compute features, classify level, score candidates, differentiate, report with confidence.
- Key thresholds: warn 26 Nm, alarm 30 Nm [P]; critical 34 Nm [I]; M 13.5 Nm watch; A6 0.4 Nm watch; persistence 2 cycles; clear margin 1 Nm.
- GEARBOX_WEAR is recognised by the three locked effects (mean, 0.75 Hz ripple, noise), TCP rise, unchanged swing and unchanged gun temperature.

17.2 Safety and escalation notes

The agent cannot act on the machine. Any recommendation to open a drive or gearbox is subject to lock-out and tag-out, the OEM procedure and site safety rules. This document is a capstone prototype and does not replace them.

17.3 References and engineering basis

- Document 01 (levels), Document 02 (signatures), Document 04 (maintenance), Document 05 (cases).

- Repository: simulation equations, MQTT contract, channelStatus.js, Phase 5 blueprint [P]; general condition-monitoring practice [E]; weights and times [I].

17.4 Revision notes

Table 42. Revision history

Rev	Date	Change
1.0	2026-10-08	Initial release.